Win Nodes for Bayes Nets: Exact Order-Statistic Queries on Gaussian Graphical Models

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Abstract

The inference stack of a Gaussian graphical model returns marginals, joint moments and samples, and none of these answers the questions a user of a correlated field most often asks: which component is the maximum, when is a threshold first crossed, who wins. We formalize these as the queries of a *win node*, an n-ary node $W = \arg\min_i X_i$ adjoined to Gaussian variables, and give exact linear-work passes for them on chains and banded precisions, where the graphical-model libraries stop at marginals. A rating model supplies the demonstration: latent ability paths with match results as evidence give a Laplace posterior that is a Gauss–Markov chain, and the peak-career query on 129,559 ATP matches returns a bimodal law for Nadal and a non-interval credible set for Agassi, neither of which any summary of the marginals contains.

1 Queries that are not marginal functionals

Let $X = (X_1, \dots, X_n)$ be jointly Gaussian: the setting of the Kalman filter, the Gaussian Markov random field, and the Gaussian belief-propagation stack. Every mainstream toolkit returns marginal moments, covariances, samples, and evidence. Four common queries are functionals of the joint law that no collection of low-dimensional marginals determines.

1. Argmax marginals $a_t = \Pr(X_t = \max_s X_s)$: which sensor reads the extreme, which index carries the peak, which entrant wins.
2. The law of the maximum $F_M(u) = \Pr(\max_t X_t \le u)$, whose complement is the excursion probability of spatial statistics.
3. First passage, the distribution of $\tau = \min\{t : X_t > u\}$: sequential monitoring, degradation to failure, barrier crossing.
4. Top-k membership $\Pr(X_i$ among the k smallest$)$: shortlist membership, portfolio inclusion, admission.

The stationary AR(1) chain makes the gap concrete. Take $X_{t+1} = \phi X_t + \sqrt{1-\phi^2}\,\epsilon_t$ with $X_1 \sim N(0,1)$: every marginal is $N(0,1)$, and a stationary Kalman filter reports exactly that at every index. The argmax marginals at $\phi = 0.9$, $n = 20$, computed by the method of Section 5 and verified against 4×10^5-path Monte Carlo to 0.007, put mass 0.091 on each endpoint and

0.035 in the middle. Interior points are squeezed by correlated-high neighbours on both sides; endpoints have one neighbour. The fact lives in the joint law, so the marginal layer cannot state it.

We call the object that answers these queries a *win node*: a deterministic n-ary node $W = \arg\min_i X_i$ (min-wins; negate for utilities) adjoined to the Gaussian variables, together with its order-statistic relatives. The definition is trivial. The observation of this paper is that for the two coupling structures that dominate practice, the node's forward query, its derivatives, and its conditioning rule are exact and cheap at the same time, which is what a graphical-model query type requires.

The ingredients are classical, and we say so once. The one-dimensional winning-time integral is the random-utility representation of Domencich and McFadden (1975); the shared survival field is competing-risks calculus; the transfer operator on a Markov chain is the sum-product algorithm; the derivative identity of Section 3 was reached for general joint laws by Müller et al. (2019). The lattice engine that evaluates and inverts these objects at scale, with its covariance grammar and tie-density Jacobians, is Cotton (2026a), on which this paper builds throughout. What is new here is the chain and band order-statistic layer with its boundary corrections, and the demonstration that a rating posterior is one of its inputs.

2 Adjacent machinery

TrueSkill (Herbrich et al., 2007) is a Bayes net with win nodes already: its factor graph attaches greater-than factors to performance differences and propagates moment-matched messages. Its win node is binary, multi-entrant events are chains of pairwise approximations, and the factor itself is approximated by expectation propagation. Section 4 gives the exact n-ary likelihood message; the TrueSkill factor is its $n = 2$ case.

Modern Gaussian message-passing systems and GMRF libraries compute marginals, free energies and samples for very large models. Their query vocabularies do not include the four functionals above. The win node is proposed as an addition to that vocabulary, not as a competitor to the toolkits.

For latent Gaussian models, Bolin and Lindgren (2015) compute excursion sets and exceedance probabilities by sequential importance sampling, and their `excursions` package is the standard tool. On chains and bands the same functionals admit the deterministic pass of Section 5. Simulation remains the recourse for general sparsity, and our software points there at its refusal boundary.

GHK evaluates one orthant probability per alternative by sequential importance sampling. Ridgway (2016) showed that on an AR(1) covariance its normalized variance grows exponentially in dimension. That is the correlated-chain case where the restricted transfer operator is exact and linear in n: high correlation is where the exact pass is tightest and the simulator is worst, and Section 6 measures the former.

3 The win node on factor-structured fields

The factor case is settled and we use it rather than develop it. The shared-field construction and its lattice discipline are Cotton (2026a, §3). Let $X = \mu + Vf + \mathrm{diag}(\sqrt{D})\,\epsilon$ with $f \sim N(0, I_r)$.

Conditional on f the components are independent, so

$$a_i \;=\; \Pr(X_i = \min_j X_j) \;=\; \mathbb{E}_f \int f_i(x) \prod_{j \neq i} S_j(x)\, dx, \tag{1}$$

with S_j the conditional survival of X_j. In English: component i wins by finishing at some value x with everyone else still above it, averaged over the shared luck. The product is shared, so building $L(x) = \sum_j \log S_j(x)$ once on a lattice of L points prices every component by the subtraction $e^{L - \log S_i}$, at $O(nLQ)$ for a Q-node quadrature.

Cotton (2026a, §5) shows the derivative to be the tie density on the face where i and j tie, so the Jacobian is minus a weighted graph Laplacian and applies in $O(nL)$ without forming an edge; Müller et al. (2019) reach the same identity for general joint laws through the surplus function. Cotton (2026b) develops this case as the posterior probability of optimality for factor-Gaussian beliefs, including the comparison against marginal-only surrogates and the bandit and stopping consequences. We take it as given here and spend the paper on the structure it does not cover.

4 Conditioning on who won

A query node becomes a Bayes-net citizen when it can be observed. Condition on $W = k$ and integrate out the factor and the winner's own noise z:

$$\Pr(W = k \mid \mu) \;=\; \mathbb{E}_{f,z} \prod_{j \neq k} \Phi\!\left(\frac{\mu_k - \mu_j + (Vf)_k - (Vf)_j + z\sqrt{D_k}}{\sqrt{D_j}} \right). \tag{2}$$

In English: given the shared luck and the winner's own performance, each rival independently had to finish behind, and each such event is one normal CDF.

The score of (2) is analytic: posterior node weights times Mills ratios, one extra pass over arrays the likelihood already computes. The message an observed win node passes to its Gaussian parents therefore carries exact first-order information. At $n = 2$ and $r = 0$ the likelihood reduces to $\Phi\big((\mu_k - \mu_j)/\sqrt{D_k + D_j}\big)$, which is TrueSkill's greater-than factor. The n-ary node needs no pairwise chaining and no moment approximation of the factor itself; moment approximations enter only if the user chooses a Gaussian posterior family, exactly as in expectation propagation. The likelihood is the multinomial probit likelihood, and it with its analytic score is the estimation core of Cotton (2026a, §5); we use it to generate a posterior rather than as a contribution. The implementation here agrees with finite differences and with dual-number automatic differentiation at 10^{-10}.

5 The order-statistic layer on chains and bands

Now let the coupling be Markov with precision bandwidth b; the Gauss–Markov chain is $b = 1$. Write the chain as $X_1 \sim N(\mu_1, \sigma_0^2)$ and $X_{t+1} \mid X_t \sim N(\mu_{t+1} + \phi_t(X_t - \mu_t), s_t^2)$. A tridiagonal precision matrix delivers these parameters directly: its Cholesky factor is bidiagonal, and each row is a conditional regression.

Max-CDF and first passage. The lattice conventions are those of Cotton (2026a, §3). Restrict the transfer operator below u. With T_t the transition kernel on the lattice and π_u the initial sub-density cut at u,

$$\Pr(\max_{s\le t} X_s \le u) \;=\; \mathbf{1}^\top \Big(\prod_{s<t} T_s^{(u)}\Big)\pi_u, \qquad T^{(u)}(x' \mid x) = T(x' \mid x)\,\mathbf{1}\{x' \le u\}, \tag{3}$$

one forward pass, $O(nL)$ per threshold with banded kernels. The survivor-mass decrements of the same pass are the first-passage law. Integrating the survival of the max over a threshold grid gives $\mathbb{E}[\max]$; the rectangle rule biases this by $O(\Delta u)$, measured at $+0.053$ on a ten-node chain, and the trapezoid removes it. A hard cell cutoff at u costs $O(\Delta x)$; assigning the boundary cell its fractional occupancy restores $O(\Delta x^2)$, which is the difference between 10^{-2} and 2×10^{-4} agreement with the i.i.d. closed form at 400 lattice points.

Argmax marginals. For each lattice threshold u_k, a forward pass restricted strictly below u_k and reaching $X_t = u_k$ gives $\alpha_t(k) = \Pr(X_1, \dots, X_{t-1} < u_k,\ X_t = u_k)$, and a backward conditional pass gives $\beta_t(k) = \Pr(X_{t+1}, \dots, X_n < u_k \mid X_t = u_k)$. Then

$$\Pr(X_t = \max_s X_s) \;=\; \sum_k \alpha_t(k)\,\beta_t(k), \tag{4}$$

integrating out the value of the maximum. With transitions precomputed the double pass is matrix slices, $O(nL^2)$ in total.

Bandwidth b by state lifting. The vector $(X_t, \dots, X_{t+b-1})$ is Markov, so the same restricted passes run on the lifted state at $O(nL^{b+1})$: linear in n, exponential only in the bandwidth. At $b = 2$ the lift covers AR(2), including the cyclical regime with complex roots that has no $b = 1$ analogue. With the second coefficient set to zero the lift reproduces the $b = 1$ pass to 5.6×10^{-16} on the same grid.

6 Numerical checks

All numbers come from committed, seeded scripts in the public repository. Monte Carlo appears only as a referee.

query	setting	exact vs. referee	cost
max-CDF	AR(1), n=50, ϕ=0.99	5×10^{-3} (MC $4{\times}10^5$)	3 ms
max-CDF	i.i.d. closed form, n=12	2×10^{-4}	—
argmax	AR(1), ϕ=0.9, n=20	7×10^{-3} (MC)	ms
argmax	random walk, n=40	4×10^{-3} (MC)	ms
band b=2	AR(2) incl. cyclical	1.1×10^{-2} (grid-limited)	230 ms/thr.
b=2 embed	ϕ_2=0 vs. b=1	5.6×10^{-16}	—
first passage	inhomogeneous chain, n=10	5×10^{-3} (MC)	$O(nL)$
$\mathbb{E}[\max]$	same	5×10^{-3} (MC band)	—

Tail thresholds. At a low threshold the max-CDF is an orthant probability, $\Pr(X_1 \le u, \dots, X_n \le u)$, and as u falls it becomes a rare event. A direct sampler needs about $1/p$ draws to see it at all, and the GHK family that computes such orthants degrades in the same regime (Ridgway, 2016). The restricted pass costs the same whatever p is: 66 ms per threshold here.

u	exact, L=400	exact, L=800	10^6 draws	draws for 10% error
0.0	6.471×10^{-3}	6.465×10^{-3}	6.432×10^{-3}	1.5×10^{4}
−1.0	1.753×10^{-5}	1.752×10^{-5}	1.60×10^{-5} (16 hits)	5.7×10^{6}
−1.5	3.067×10^{-7}	3.079×10^{-7}	no hits	3.2×10^{8}
−2.0	2.520×10^{-9}	2.521×10^{-9}	no hits	4.0×10^{10}
−2.5	9.380×10^{-12}	9.418×10^{-12}	no hits	1.1×10^{13}
−3.0	1.576×10^{-14}	1.578×10^{-14}	no hits	6.3×10^{15}

Stationary AR(1), $n = 50$, $\phi = 0.9$. Simulation agrees where it has hits and returns zero where it does not, which is an absence of information rather than a small error. Below $u = -1.5$ no referee is available at all, and the evidence for the exact values is that they are stable under grid refinement: the two passes agree to three or four significant figures at every threshold, including 1.58×10^{-14}.

Two consequences follow from the argmax table rather than merely illustrating it. The boundary pile-up law is one: positive serial correlation pushes the argmax of a stationary path to the endpoints, 2.6× at $\phi = 0.9$, and no marginal summary can express this. The finite-n arcsine refinement is the other. On a random walk the transfer operator matches the finite-n argmax law to 0.3% while the continuum arcsine density sits 1% away, because the engine computes the exact finite law of which arcsine is the limit. The same pass handles the drifted, barriered and mean-reverting cases that have no closed form. The accuracy profile is complementary to simulation: the exact pass is tightest at high correlation, exactly where Ridgway (2016) shows the GHK family degrading.

7 Peak seasons from win evidence

A rating model is a Bayes net whose evidence is wins, and it produces exactly the posterior this paper queries. Give one player a latent ability path $\theta_1, \ldots, \theta_T$ across the years of a career and put a random-walk prior on it. Each match against an opponent of ability a_o contributes $\log \Phi(\pm(\theta_t - a_o))$ by (2) at $n = 2$, a term touching one year, so the likelihood's Hessian is diagonal. The prior's precision is tridiagonal, the Laplace posterior precision is their sum, and the career posterior is therefore a Gauss–Markov chain of the kind Section 5 reads.

The data are 129,559 ATP matches from 1985 to 2024 and the 1,453 players with at least fifteen of them. Static abilities for the field come from a MAP fit of the probit Bradley–Terry likelihood. One player at a time then has that constant replaced by a yearly path, fitted under the random-walk prior with the innovation scale chosen by the Laplace marginal likelihood. Opponents stay at their static strength, which is a conditioning step rather than a joint fit.

Reading the peak off the fitted path is what a rating table invites. The argmax marginals answer the question that was asked.

player	fitted peak	Pr(that year)	80% credible set
Federer	2006	0.52	2005–2006
Nadal	2013	0.27	six years within 2008–2019
Djokovic	2015	0.43	2013–2015
Sampras	1994	0.47	1994–1996
Agassi	1995	0.55	1995, 2002–2003
Murray	2016	0.59	three years within 2009–2016

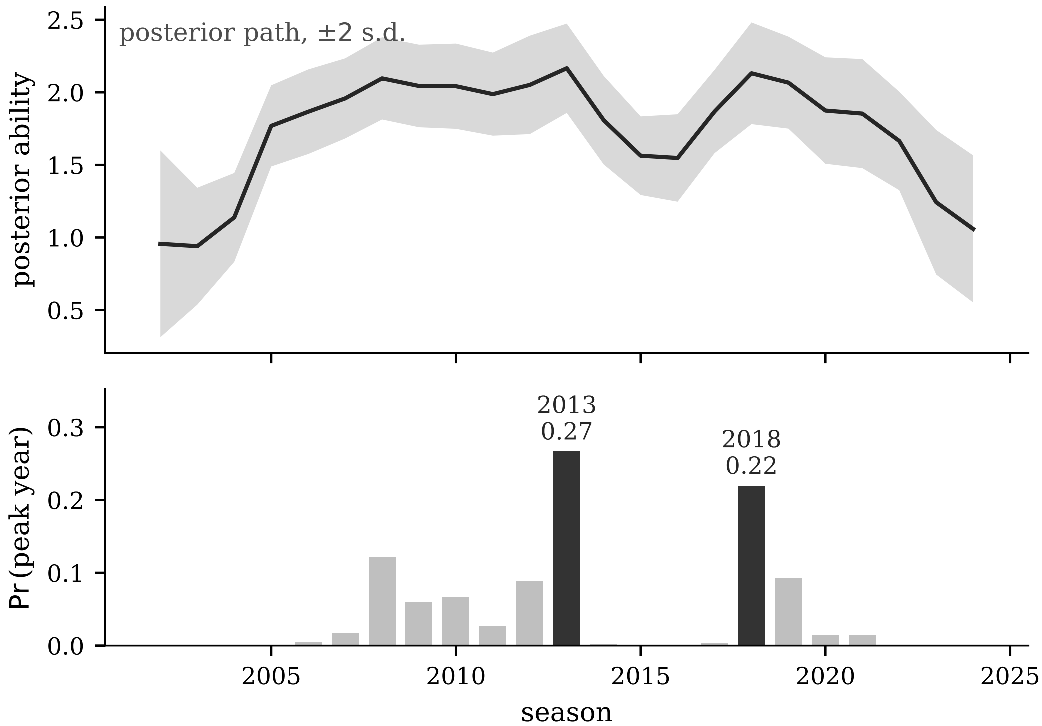

Figure 1: Rafael Nadal's posterior ability path from 1,317 matches (top) and the law of its argmax (bottom). The path is one broad plateau and the argmax law is bimodal, at 2013 and 2018, with the seasons between them carrying almost no mass. The band in the top panel does not contain that statement, and no summary of the yearly marginals does either.

The point answers are the ones a reader would expect, and stating them alone would overclaim. Federer's 2006 carries 0.52 against 0.41 for 2005. Nadal's most likely peak year carries 0.27, and his law is bimodal: 0.267 at 2013 and 0.220 at 2018, with the years between them small. Agassi's credible set is not an interval. It holds 1995 and the 2002–2003 resurgence and excludes the years in between, which is a statement about the joint law and not about any year's marginal.

Figure 1 puts the two objects on one time axis. The mechanism behind the second feature is visible in the same fit. At equal posterior mean, the year with fewer matches carries more argmax mass, because fewer matches leave a wider posterior and a wider posterior has more room to hold the maximum. Nadal's 2010 has posterior mean 2.042 from 80 matches and probability 0.066; his 2012 has mean 2.051 from 48 matches and probability 0.088. A table of yearly ratings with error bars does not contain that comparison, because the comparison is between joint events.

Two hundred thousand draws from each posterior referee the exact pass, with maximum discrepancy 0.0019 across all six careers and every year. The Gaussian posterior is approximate, as it is in every rating system that reports one. The query on that posterior is not.

8 Software

The factor win node, its top-k relatives, both Jacobian blocks and the calibration inverses ship in the `winning` package of Cotton (2026a): a Python reference with R, JavaScript, Julia and Rust ports pinned to embedded parity vectors, most at 10^{-15}. The observed win node, likelihood (2) with analytic score, is the estimation core of `MultinomialProbit.jl`, whose score is additionally refereed by forward-mode automatic differentiation through the same code path. The chain and band layer is `GMRFExtremes.jl`, which reads a Gauss–Markov chain off a tridiagonal precision and dispatches on GMRF library types. Wider bandwidth and general sparsity are refused with a pointer to the lifted pass and to Bolin and Lindgren (2015) respectively.

9 Scope

The reduction requires structure. Factor coupling of rank r costs a Q-node quadrature and is exact to quadrature accuracy; precision bandwidth b costs $O(nL^{b+1})$ and is practical for $b \leq 3$; general sparse precision gets neither treatment here, and the excursion functionals then belong to the sampling methods of Bolin and Lindgren (2015). The win-node concept itself is not new: TrueSkill carries a binary one, Müller et al. (2019) derived the tie-density derivative for general laws, the lattice engine and its Jacobians are Cotton (2026a), and the factor query is Cotton (2026b). Equation (2) is the multinomial probit likelihood and is classical. New here are the restricted-pass order-statistic layer for chains and bands, with its boundary corrections and state lift, and the peak-career result of Section 7. We leave to future work the lifted-state argmax at $b \geq 2$ and message compression past $b \approx 3$.

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